EMAIL SECURITY ANALYSIS REPORT

PROJECT : 02
ASSESSMENT TYPE : ANALYSING EMAILS
DATE : 24-02-2026

EXECUTIVE SUMMARY

Phishing is one of the most common cyber threats targeting businesses and individuals. Attackers attempt to trick users into clicking malicious links, downloading infected attachments, or sharing sensitive information such as passwords and OTPs.

This report analyzes three real-world email samples and classifies them as:

- 🟢 Safe
- 🟡 Suspicious
- 🟥 Phishing

The goal is to demonstrate phishing detection techniques and provide awareness guidelines for users.

TOOLS USED

The following tools were used during the investigation:

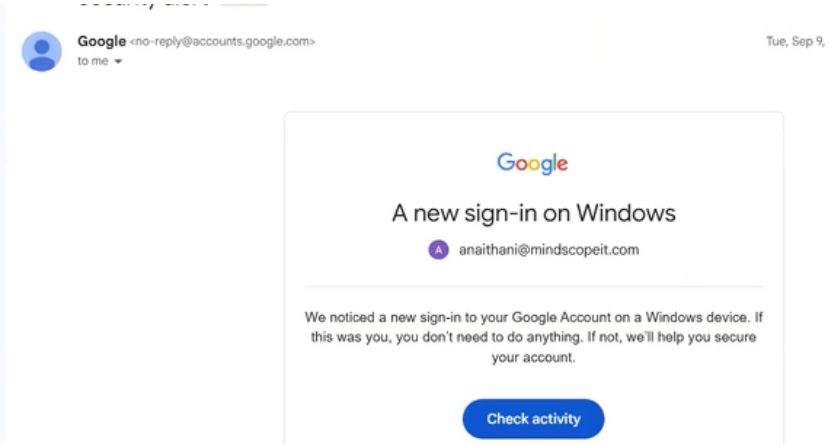
- Google Message Header Analyzer
- MXToolbox (Email Header & Domain Analysis)
- VirusTotal (Domain Reputation Check)
- Gmail Header Inspection
- Browser Domain Inspection

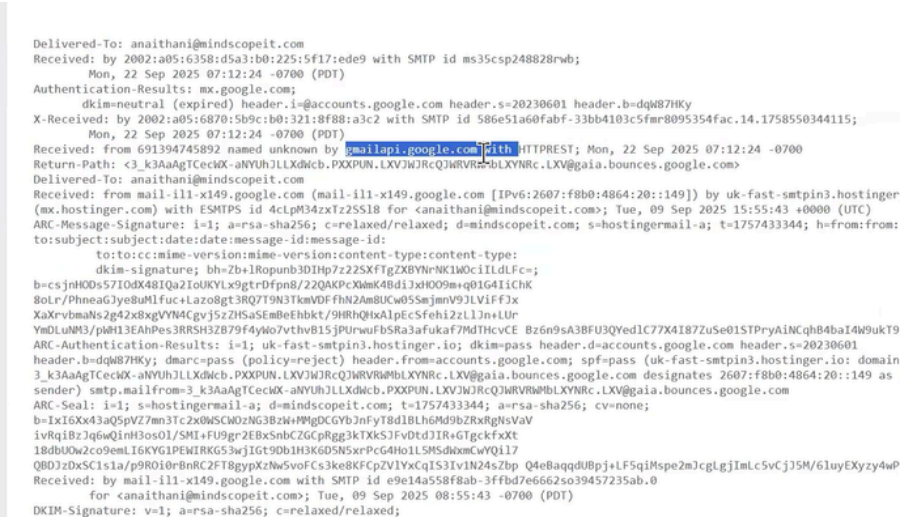
These tools helped verify sender authentication, inspect domains, and evaluate potential threats.

EMAIL ANALYSIS & CLASSIFICATION

Email 1: Google Security Alert – “New Sign-in on Windows”

Classification: SAFE





```
Delivered-To: anaithani@mindscopeit.com
Received: by 2002:a05:6358:d5a3:b0:225:5f17:ede9 with SMTP id ms35csp248828rwb;
        Mon, 22 Sep 2025 07:12:24 -0700 (PDT)
Authentication-Results: mx.google.com;
       dkim=neutral (expired) header.i=@accounts.google.com header.s=20230601 header.b=dqW87HKy
X-Received: by 2002:a05:6870:5b9c:b0:321:8f88:a3c2 with SMTP id 586e51a60fabf-33bb4103c5fmr8095354fac.14.1758550344115;
        Mon, 22 Sep 2025 07:12:24 -0700 (PDT)
Received: from 691394745892 named unknown by gmailapi.google.com with HTTPREST; Mon, 22 Sep 2025 07:12:24 -0700
Return-Path: <3_k3AaAgTCecWX-aNYUhJLLXdWcb.PXXPUN.LXVJWJRcQJWRVRWMbLXYNRc.LXV@gaia.bounces.google.com>
Delivered-To: anaithani@mindscopeit.com
Received: from mail-il1-x149.google.com (mail-il1-x149.google.com [IPv6:2607:f8b0:4864:20::149]) by uk-fast-smtpin3.hostinger
(mx.hostinger.com) with ESMTPS id 4cLpM34zxTz2SSl8 for <anaithani@mindscopeit.com>; Tue, 09 Sep 2025 15:55:43 +0000 (UTC)
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=mindscopeit.com; s=hostingermail-a; t=1757433344; h=from:from:
to:subject:subject:date:date:message-id:message-id:
        to:to:cc:mime-version:mime-version:content-type:content-type:
        dkim-signature; bh=Zb+lRopunb3DIHp7z22SXfTgZXBYNrNK1WOciILdLFc=;
b=csjnHODs57IOdX48IQa2IoUKYLx9gtrDfpn8/22QAKPcXWmK4BdiJxHOO9m+q01G4IiChK
8oLr/PhneaGJye8uMlfuc+Lazo8gt3RQ7T9N3TkmVDFfhN2Am8UCw05SmjmnV9JLViFfJx
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ARC-Authentication-Results: i=1; uk-fast-smtpin3.hostinger.io; dkim=pass header.d=accounts.google.com header.s=20230601
header.b=dqW87HKy; dmarc=pass (policy=reject) header.from=accounts.google.com; spf=pass (uk-fast-smtpin3.hostinger.io: domain
3_k3AaAgTCecWX-aNYUhJLLXdWcb.PXXPUN.LXVJWJRcQJWRVRWMbLXYNRc.LXV@gaia.bounces.google.com designates 2607:f8b0:4864:20::149 as
sender) smtp.mailfrom=3_k3AaAgTCecWX-aNYUhJLLXdWcb.PXXPUN.LXVJWJRcQJWRVRWMbLXYNRc.LXV@gaia.bounces.google.com
ARC-Seal: i=1; s=hostingermail-a; d=mindscopeit.com; t=1757433344; a=rsa-sha256; cv=none;
b=IxI6Xx43aQ5pVZ7mn3Tc2x0WSCWOzNG3BzW+MMgDCGYbJnFyT8dlBLh6Md9bZRxRgNsVaV
ivRqiBzJq6wQinH3osOl/SMI+FU9gr2EBxSnbCZGCpRgg3kTXkSJFvDtdJIR+GTgckfxXt
18dbUOw2co9emLI6KYG1PEWIRKG53wjIGt9Db1H3K6D5N5xrPcG4Ho1L5MSdWxmCwYQil7
QBDJzDxSC1s1a/p9ROi0r8nRC2FT8gypXzNw5vofCs3ke8KFCpZVlYxCqIS3Iv1N24sZbp Q4eBaqqdUBpj+LF5qiMspe2mJcgLgjImLc5vCjJ5M/6luyEXyzy4wP
Received: by mail-il1-x149.google.com with SMTP id e9e14a558f8ab-3ffbd7e6662so39457235ab.0
        for <anaithani@mindscopeit.com>; Tue, 09 Sep 2025 08:55:43 -0700 (PDT)
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
```

Analysis:

- Sender: no-reply@accounts.google.com
- SPF: PASS
- DKIM: PASS
- DMARC: PASS (policy=reject)
- Return-path aligned with Google infrastructure

The email content was consistent with legitimate Google security alerts. No malicious links, attachments, or credential-harvesting attempts were identified.

Conclusion:

The email was authenticated and originated from legitimate Google servers. No indicators of compromise were found. Classified as Safe.

🟡 Email 2: “Republic Day Offer Ending Today” – HackingFlix

Classification: SUSPICIOUS

🔴 Expiring Today, Fathima sithara Spam ×

Gautam Kumawat <support@hackingflix.com>
to me

Hey Fathima sithara,

🇮🇳 **Republic Day Offer Ending Today** 🇮🇳

Last Hours Left to Grab the Republic Day Offer on Certified Cyber Warrior.

98% OFF + ₹1000 OFF Republic Day Discount
Lifetime Access + Lifetime Support + Global Certification

So **many students have enrolled with us this time** and have taken the right step in the right direction for their learning!

This is your LAST CHANCE, Fathima sithara!

Image item

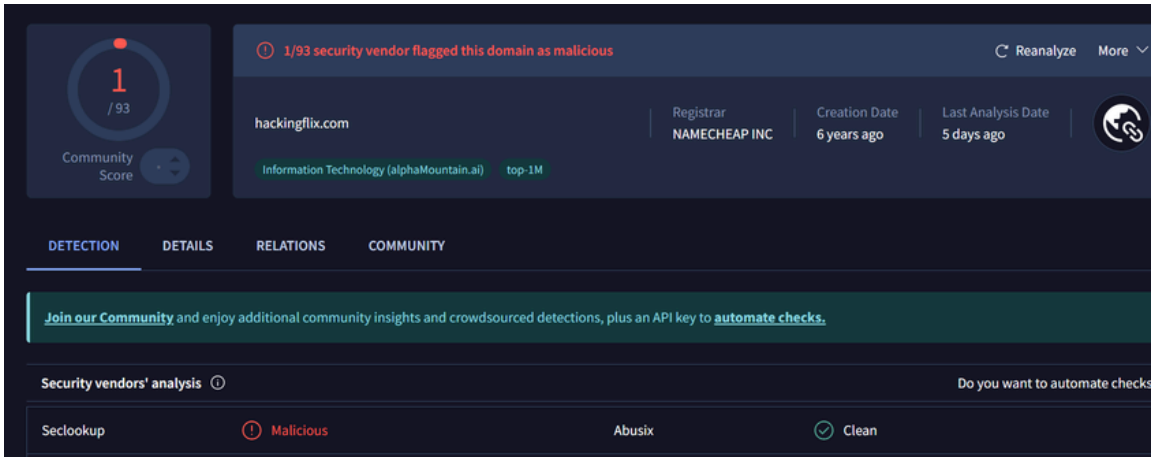
Grab this chance now before it ends!

To Your Success,
Gautam Kumawat
Founder, HackingFlix

Original Message

Message ID	<j3gbUa4vQu-IKMXpQwQmUw@geopod-ismtpd-32>
Created at:	Sun, Feb 1, 2026 at 7:46 PM (Delivered after 2 seconds)
From:	Gautam Kumawat <support@hackingflix.com>
To:	[redacted]
Subject:	Expiring Today, Fathima sithara
SPF:	PASS with IP 149.72.58.57 Learn more
DKIM:	'PASS' with domain hackingflix.com Learn more
DMARC:	'PASS' Learn more

- DMARC Compliant
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated



Analysis:

- Sender: support@hackingflix.com
- SPF: PASS
- DKIM: PASS
- DMARC: PASS
- VirusTotal: 1/93 vendors flagged domain

Indicators observed:

- Strong urgency ("LAST CHANCE")
- Extreme discount (98% OFF)
- Promotional pressure tactics
- Unsolicited marketing content

No impersonation or credential harvesting was identified.

Conclusion:

Although authentication passed and no confirmed malicious behavior was detected, aggressive marketing and urgency tactics make this email Suspicious.

Email 3: “May Supplement Order 04/05” – hardroot.biz

Classification: PHISHING

Wed 5/4/2022 9:37 PM

Emily Yen <doc@hardroot.biz>

May Supplement Order 04/05

To undisclosed-recipients:

Follow up.
This message has been replied to or forwarded.

Supplimentary Order 04 MAY.xlsx
206 KB

Dear Customer,

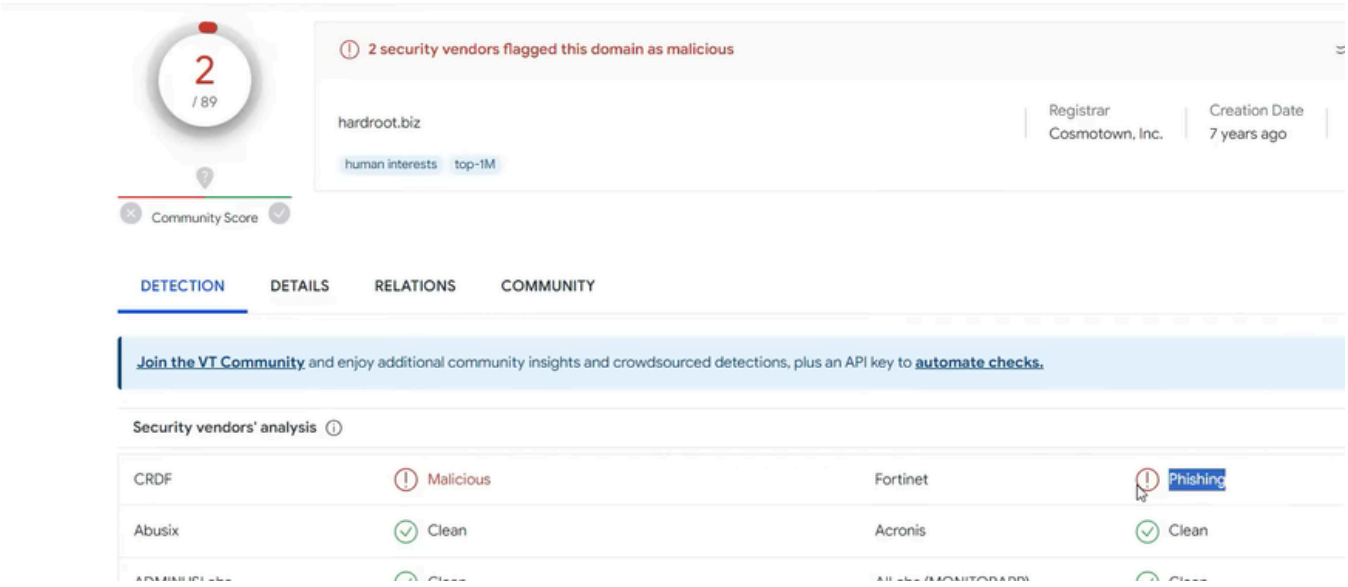
Please acknowledge our request as attached.

Andrew
0100|04

Headers Found

Header Name	Header Value
From	"Emily Ye[illegible]" <doc@hardroot.biz>
To	<undisclosed-recipients:>
Subject	May Supplement Order 04/05
Date	Wed, 4 May 2022 21:37:09 +0530
Message-ID	<3b09905aaa0c1813e5725c294f5dc5a8@lawaholdings.co.bw>
MIME-Version	1.0
Content-Type	multipart/mixed; boundary="----=_NextPart_000_0010_01D8C3AE.70560100"
X-Mailer	Microsoft Outlook 16.0
X-CMAE-Analysis	v=2.4 cv=YvvK+6UX c=1 sm=1 tr=0 ts=6272a731 cx=a_idp_d a=Dj3c+FHrcchNxiG6n/fo3A==:117 a=Dj3c+FHrcchNxiG6n/fo3A==:17 a=oZkIemNP1mAA:10 a=Q02_sXRfmUlnSk7_BrYA:9 a=CjuIK1q_8ugA:10 a=NkTa0eAkhXMAxvwh2FwA:9 a=fMZ_E1CIlfQA:10 a=IC_P5zELcC8A:10 a=k92MKxcbr7QA:10 a=gKT_8TCvE1r4shHETz0G:22 a=Z5ABNNGmrOfJ6cZ5blyy:22 a=QOGEsqRv6VhmHaoFNykA:22
X-CMAE-Envelope	MS4xfNsETP1QpJUApYqq1fbCwciM0c75kKj6VrC9rVb1q0d+BfZPvzqonyyjozSZDrvP2wt2u1TUQVi2hBJTYjoRVy9dh9FpAHCVgEOmdTzeZeYb8qYs+VL/ kbfWHvilzSqPm+DRdFOaTpoT+4DRKohDGoyTyaH4TMtC+hPSQ7I8Q23XHuJKlaA8aWqfPS9pm0eJgg==
Thread-Index	AQKzUibJnXF8xTSqwnuzicahp+uesQ==
X-Sender	doc@hardroot.biz
X-OlkEid	00000000D6AB40BA601F9742A4C86F3FC35C047D0700178FD8DA7507254D88C87DBF448EE1D2000000000221000178FD8DA7507254D88C87DBF448EE1D2000000001C00000081A0C022449BF34D8F38B80C88009984

- DMARC Compliant (No DMARC Record Found)
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated



Analysis:

- Sender: doc@hardroot.biz
- Attachment: Supplementary Order 04 MAY.xlsx
- Sent to undisclosed recipients
- Generic greeting ("Dear Customer")
- Unexpected invoice-style attachment
- Unknown sender domain

Invoice-themed Excel attachments are commonly used to distribute malware via malicious macros.

Indicators identified:

✓ Unknown domain
✓ Attachment-based delivery
✓ Generic greeting
✓ No prior business relationship
✓ Social engineering tactic (invoice lure)

Conclusion:

Due to malicious attachment risk and phishing indicators, this email is classified as Phishing.

HOW PHISHING ATTACKS WORK

Phishing attacks rely on social engineering rather than technical hacking.
Attackers:

- Create fake urgency (account suspension, last chance offer)
- Impersonate trusted organizations
- Send malicious links or infected attachments
- Trick users into entering credentials or downloading malware

Most phishing attacks succeed because users trust the message and act quickly without verifying it.

COMMON PHISHING INDICATORS

Employees should watch for:

✓ Spoofed or unfamiliar sender domains

✓ Generic greetings (“Dear Customer”)

✓ Urgent or fear-based language

✓ Unexpected attachments (especially .xlsx, .zip, .html)

✓ Requests for passwords, OTPs, or confidential data

✓ Suspicious or shortened links

PREVENTION GUIDELINES

To reduce phishing risk:

- Always verify sender email domains carefully
- Do not open unexpected attachments
- Hover over links before clicking
- Never share passwords or OTPs via email
- Report suspicious emails to IT/security team
- Use multi-factor authentication (MFA)
- Keep systems and antivirus updated

DO'S AND DON'TS FOR EMPLOYEES

DO'S

- Verify email authenticity before acting
- Check header authentication if unsure
- Confirm financial or invoice emails via phone
- Report suspicious emails immediately

DONT'S

- Click links from unknown senders
- Download unexpected attachments
- Share login credentials via email
- Panic due to urgency tactics

FINAL RISK SUMMARY

Email	Classification	Risk Level
Google Security Alert	Safe	Low
HackingFlix Offer	Suspicious	Medium
hardroot.biz Invoice	Phishing	High

The analysis highlights how email threats vary in risk level and intent. While some emails are legitimate, others use urgency, attachments, or social engineering tactics to manipulate users. Proper verification and user awareness remain the strongest defense against phishing attacks.

CONCLUSION

This investigation demonstrates how phishing emails can be identified through:

- Header authentication analysis
- Domain inspection
- Content-based red flag detection
- Attachment risk evaluation

By combining technical verification with user awareness training, organizations can significantly reduce phishing-related incidents.

Phishing prevention depends not only on secure systems but also on informed and vigilant users.